

SUPPLY CHAIN PERFORMANCE INTELLIGENCE

Project 14 | SQL + Power BI | Engineers' Day Hackathon 2026

Project Overview

This project uses the DataCo Smart Supply Chain Dataset to analyze supply-chain performance through MySQL and Power BI. The analysis focuses on delivery performance, actual versus scheduled shipping time, shipping-mode performance, regional performance, product performance, sales, profitability and late-delivery risk.

Business Problem

A manufacturing company has experienced increasing delivery delays and logistics challenges. Management needs a single intelligence layer to understand supply-chain performance and identify the areas contributing to delivery problems.

Objective

To analyze supply-chain performance using SQL and Power BI and identify products, shipping modes and regions contributing to delivery delays and operational performance issues.

Technology Used

- MySQL
- Power BI
- Power Query
- DAX
- GitHub

Data Preparation

- Raw CSV loaded into a MySQL staging table.
- Data cleaned and transformed using trimming, NULL handling, type conversion and date conversion.
- Created **fact_supply_chain**.
- Created **dim_product**, **dim_location**, **dim_shipping** and **dim_date**.

SQL Analysis

- Delivery-status counts
- Shipping-mode performance
- Regional orders, sales and profit
- Top products by orders and profit
- Average delivery delay by shipping mode
- Products with highest late-delivery rates
- Regions with highest delivery delays
- Sales and profit by shipping mode

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Power BI Dashboard

- Home
- Executive Overview
- Problem Diagnosis
- Action and Recommendations

Key Findings

- Overall late-delivery rate stands at 55%, confirming late delivery as a major operational issue.
- Second Class shipping shows the largest gap between actual and scheduled shipping time — 4 days actual against 2 days scheduled, a 2-day delay.
- Regional differences in delivery performance are identified, with Central Africa, East Africa and South Asia showing the highest late-delivery rates.
- Products such as the SOLE E25 Elliptical and Ogio Race Golf Shoes show late-delivery rates above 65%, making them priority candidates for review.
- Shipping modes are compared using volume, sales, profit and delay, with Second Class showing the widest actual-vs-scheduled gap among all modes.

Recommendations

- Prioritize regions with consistently high late-delivery rates, particularly Central Africa and East Africa.
- Review Second Class shipping specifically, where actual delivery time exceeds the scheduled time by 2 days.
- Investigate high late-delivery products such as the SOLE E25 Elliptical and Ogio Race Golf Shoes to identify root causes.
- Use sales, profit and delivery performance together when setting priorities, so fixes target regions and products that matter most to the business.

Conclusion

The project creates a practical supply-chain intelligence workflow from raw order-level data to SQL analysis and Power BI reporting. It helps decision-makers understand delivery performance, compare shipping modes and regions, identify higher-risk products, and decide where further operational investigation should begin.